

Harvard University
Physics 141 Pset 6

Due November 14, 2025, 5 PM

1. **Introduction to the delta function:** The delta (generalized) function can be defined as

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases}$$

such that

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

- (a) First, use integration by parts to prove that

$$\int f(x) \delta(x) dx = f(0)$$

for any function f which tends to zero at positive and negative infinity.

- (b) We defined a forward Fourier transform of some function $f(t)$ to be

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

and the inverse Fourier transform as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega t} d\omega$$

Show that the Fourier transform of the delta function is 1.

- (c) Express the x -translated delta function $\delta(x - \mu)$ as a complex integral.
- (d) Now, we will show an example of how this form of the delta function can be useful. Let a particle in 1D perform a random walk with each step pulled from a gaussian with mean μ and standard deviation σ . Then, write the first two steps as x_1 and x_2 . First, argue that

$$\text{prob}(x_1 + x_2 = k) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x_1 + x_2 = k) p(x_1) p(x_2) dx_1 dx_2$$

where $p(x)$ is the probability of drawing some x from the gaussian with mean μ and standard deviation sigma.

- (e) Now, plug in $p(x)$ and the integral form for $\delta(x)$ to find the distribution of the endpoint of the random walker after two steps.
- (f) Explain why your answer is a gaussian with mean zero and standard deviation $\sqrt{2}\sigma$. What would you expect it to be for an arbitrary number N steps?
- (g) [Extra credit] Prove that the gaussian distribution of the endpoint of the random walk holds for an arbitrary number N steps.
- (h) Prove that the Fourier transform of a Gaussian of mean 0 and standard deviation σ is a Gaussian. What happens to the mean and standard deviation? Is the Fourier transform normalized?
- (i) What happens when you take the limit $\sigma \rightarrow 0$ of the standard deviation of a gaussian? Argue that this is a useful representation of the delta function by examining the Fourier transform.

2. In this problem, you will explore Fourier transforms and study how biological systems apply filters to stimuli. To start, let's define some notation for the continuous case.

- Let $x(t)$ and $h(t)$ denote two continuous time-series.
- Let $X(\omega)$ and $H(\omega)$ denote their Fourier transforms. To be clear, we can write

$$X(\omega) = \mathcal{F}(x)(\omega) = \int_{-\infty}^{\infty} dt x(t) e^{-i\omega t}$$

- Define the convolution of two continuous time-series as

$$(x * h)(t) = \int_{-\infty}^{\infty} d\tau x(\tau) h(t - \tau)$$

- (a) Prove the Fourier convolution theorem. That is, show that if $y = (x * h)$, then $Y(\omega) = X(\omega)H(\omega)$.

Some receptor/signaling pathways that bacteria use to detect ligands act as low-pass filters. In an experiment, when you measure the response of bacteria to a ligand time course, your measurement will be a convolution of the true ligand time course with an impulse response. Model the impulse response $h(t)$ with an exponential function

$$h(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} u(t)$$

Where $\tau > 0$ and $u(t)$ is the Heaviside step function ($u(t) = 1$ for $t \geq 0$ and 0 otherwise). Let us model the true ligand time course with a sinusoid

$$x(t) = A \cos(\omega_0 t)$$

Where $A > 0$. The bacteria sensing module is driven with a sinusoid of frequency ω_0 . As we stated above, the output $y(t)$ (what we measure) is given by the convolution

$$y(t) = (x * h)(t)$$

- (b) Show that the frequency response $H(\omega)$ is given by

$$H(\omega) = \frac{1}{1 + i\omega\tau}$$

- (c) Show that $x(t)$ has the Fourier transform

$$X(\omega) = A\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$$

- (d) Using the convolution theorem that you proved in (a), find $Y(\omega)$.

- (e) By taking an inverse Fourier transform, show that

$$y(t) = A\Re(H(\omega_0)e^{i\omega_0 t})$$

Note that $H(-\omega_0) = (H(\omega_0))^*$ (* denotes complex conjugation). Note that $y(t)$ is a real number. A trick to solving this problem is to use the fact that for a complex number c ,

$$c + c^* = 2\Re(c)$$

Where $\Re$ denotes taking the real part (i.e. if $c = a + bi$, $\Re(c) = a$).

- (f) Define the gain (the amplitude response) as $G(\omega) = |H(\omega)| = \sqrt{H(\omega)(H(\omega))^*}$, where $(H(\omega))^*$ denotes the complex conjugate of $H(\omega)$. Find the gain G .
- (g) Define the phase $\phi(\omega)$ as the argument of $H(\omega)$. Find the phase ϕ .
- (h) Rewrite $y(t)$ in terms of the gain $G(\omega_0)$ and phase $\phi(\omega_0)$. You should be able to show that

$$y(t) = AG \cos(\omega_0 t + \phi)$$

- (i) In the limits of $\omega_0 \ll \frac{1}{\tau}$ and $\omega_0 \gg \frac{1}{\tau}$, what is the gain and phase? Provide a biological interpretation of this result (how does your measurement/the receptor track fast/slow changes in the ligand concentration).